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Research Article

New Coating TENG with Antiwear and Healing Functions for Energy Harvesting

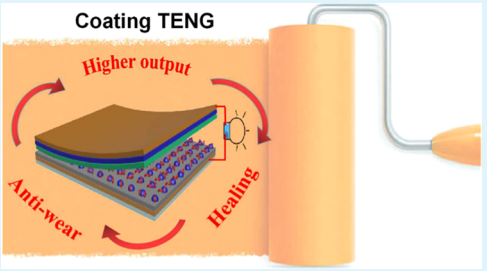
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ABSTRACT: In view of the limitations of practical applications of current triboelectric nanogenerators (TENGs), a new type of coating TENGs with antiwear and healing properties have been fabricated to collect the large-scale dissipative energy in the environment. To enhance the triboelectrification performance of the coating TENG, mesoporous silica filled with perfluorooctylethanol is added to the acrylate resin material, in addition to improving the antiwear properties of the frictional coating. The result shows that when the mesoporous silica is used as a carrier and perfluorooctylethanol is loaded, the short-circuit current (I_{sc}) and output voltage (V_o) of the coating TENG reach as high as 10 μA and 220 V, respectively, which are 4−5 times higher than those of pure acrylate. More importantly, the coefficient of friction of the new coating decreases from 0.11 to 0.04 with the wear volume reducing by approximately 89%, indicating a better friction-reducing property of the coating for long-term working. As a new coating material based on the traditional acrylic resin, it can be widely sprayed onto various walls, metals, and hulls as protection coating as well as power-generation coating. Interestingly, when the coating is damaged due to long-term aging or external mechanical forces, it can restore its triboelectric performance by encapsulating the repair agent within the pore structure of silica. Owing to the large-area fabrication, low cost, high output performance, and antiwear properties, the new coating TENGs have promising potential for practical applications in energy-harvesting, self-energy supplies, and self-powered sensors.



KEYWORDS: *coating TENG, mesoporous silica, perfluorooctylethanol, antiwear and friction reducing, healable*

1. INTRODUCTION

Energy is one of the pillars of modern society. Traditional energy can no longer keep up with rapid developments in science and technology. Thus, a renewable energy source to replace traditional energy sources should be found. Collectable environmental mechanical energy has become a key research issue in the field of energy harvesting.[1−3] A new type of energy-harvesting device called a triboelectric nanogenerator (TENG) has been widely and rapidly developed due to its simple design structure, low cost, high conversion efficiency, large friction energy, and environmental friendliness.[4,5] The working principle of the triboelectric nanogenerator (TENG) is based on the coupling effects of triboelectrification and electrostatic induction.[6−8] Two friction layers rub against each other and generate a friction potential difference to drive the electron flow in the external circuit. In this way, TENG could collect almost all kinds of environmental mechanical energy into electrical energy in daily life. In other words, TENG can collect various forms of mechanical energy from mechanical vibration,[9,10] human motion,[11] wind,[12,13] sound, and ocean waves.[14−16] The new generator can also be used as a power source to drive electronic devices, such as wearables, detectors, and sensors,[17,18] However, most of the TENGs are microdevices and the small friction current and inability to be used on a large scale limit the practical applications of these devices. Therefore, finding a material that can increase its current output and technology that can be applied for large-scale fabrication is an important direction.

At present, coating technology is being applied in various aspects of life.[19−21] The development and application of surface coating technology provide superior performance to the matrix material, and the large-scale application of TENGs depends on the coating technology. However, surface coating technology also has limitations, such as low wear and abrasion resistance.[22−27] Therefore, the main content of the current is applied to the surface coating technology to achieve the large-scale production of TENGs while overcoming the defects of the coating itself. For example, acrylate is a common coating used in the industrial production of automobiles, aircraft, medical equipment, instrumentation, and wood furniture due to its advantages of high strength, high heat resistance, and good film formation.[28−30] However, acrylate coatings have defects that need to be addressed urgently, such as low hardness, easy

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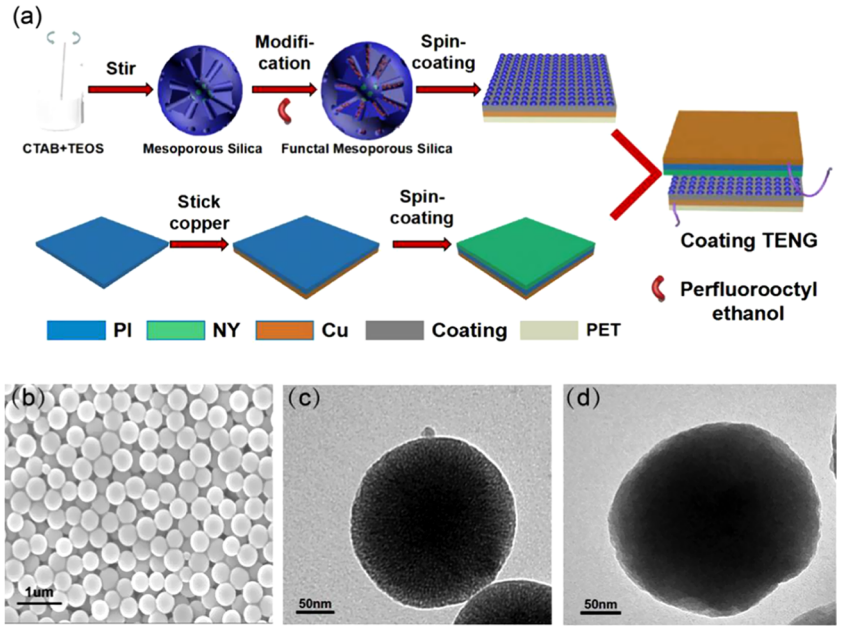


Figure 1. (a) Scheme of the fabrication processes of coating triboelectric nanogenerator with mesoporous silica-loaded perfluorooctylethanol. SEM (b) and TEM (c, d) images of mesoporous silica before and after filling perfluorooctylethanol.

breakage, poor tribological properties during solid−solid friction, and difficulty in generating micro/nanostructures to improve power generation efficiency. Therefore, it is better that the coating TENG should be modified in a manner that provides antiwear, friction-reducing, and healing functions.

In this study, we combined coating and TENG technology to solve the problem of large-scale application of TENGs and improved the triboelectric output and wear resistance by adding a filler to the coating. Owing to characteristics such as small particle size, large specific surface area, and superior hydrophilicity, stability, thickening, and prevention, SiO_2 particles are widely used in the coating industry to improve antiwear and friction-reducing properties.[24−27] Here, we filled the nano-SiO_2 particles into the acrylate coating to improve the wear resistance and extend the working durability. To further improve the output of TENGs, we introduced modifier molecules with a negative friction polarity (especially the composite materials containing F and high electron affinity) into the coating to build a high-performance TENG. In this paper, we selected the mesoporous silica as the wear-resistant additive material and also micro- and nanocontainer to load the addition of perfluorooctylethanol. On the other hand, to solve the problem of damage to surface coating materials in the actual application process, we studied the process of postheat treatments after a surface coating material with perfluorooctylethanol was damaged. We detected that the TENG output decreased after the removal of perfluorooctylethanol on the SiO_2 surface by O_2 plasma and then increased after heat repairing in an oven, indicating a healing function of the new coating TENG for long-term working.

2. EXPERIMENTAL SECTION

2.1. Materials. Acrylate (SK6303) was obtained from Shanghai Shuaike Chemical, 1*H*,1*H*,2*H*,2*H*-perfluoro-1-decanol was bought from Hubei Jusheng Technology, and nylon 11 (NY 11, 98%, piglet) was acquired from Tianjin Heowns Biochemical Technology. Polyimide (PI) films, poly(ethylene terephthalate) (PET) films, and Cu tape were purchased from a local market. All of the other chemicals were of analytical grade and used without further treatment.

2.2. Mesoporous Silica Synthesis. The mesoporous silica was prepared through the NH_3-catalyzed hydrolysis of tetraethyl orthosilicate (TEOS) in mixed water−ethanol solvents with hexadecyl trimethyl ammonium bromide (CTAB) as a template. First, 100 mL of deionized water, 10 mL of NH_3 water, and 55 mL of ethanol were mixed and stirred for 30 min. Then, 0.06 mol CTAB was added and vigorously stirred for 30 min to obtain a clear solution. Second, TEOS was slowly dropped into the solution and stirred for 3 h. Thereafter, the mixture was allowed to stand under static conditions at room temperature for 12 h, thereby leading to the formation of a white precipitate. Finally, the white precipitate was filtered, washed, and calcined in the air at 600 °C for 2 h to obtain the mesoporous silica.

2.3. Perfluorooctylethanol-Loaded Mesoporous Silica (F-SiO_2) Preparation. Two grams of 1*H*,1*H*,2*H*,2*H*-perfluoro-1-decanol was added to 40 mL of *N*,*N*-dimethylformamide solution. After uniform stirring, a certain amount of the sample in the mesoporous silica particles was added so that the mass fraction was 5% and the mixture was ultrasonically shaken for 30 min to remove the gas. The mixture was stirred for another 12 h and then filtered, washed, and air-dried at room temperature to obtain F-SiO_2.

2.4. Preparations of Coating Triboelectrode Material Containing F-SiO_2 and TENG. The preparation process of the TENG mixed with mesoporous silica is shown in Figure 1. First, a Cu tape was adhered to one side of the PET film and a Cu wire was adhered to the other side of the Cu foil to form a preliminary electrode. Second, the coating solution was prepared. A certain mass fraction of mesoporous silica particles was mixed with 2 g of DMF solution and ultrasonically shaken for 10 min to uniformly disperse the silica particles. Then, the DMF/SiO_2 mixture was added to the acrylate coating at a certain proportion and stirred for 10 min to produce a mixed coating. Thereafter, the coating and nylon 11 solutions were spin-coated onto a 4 × 4 cm^2 Cu/PET film with the rotation speed of 3000 rpm for 30 s to form a friction layer electrode. The coating and nylon 11 solutions were dried in an oven at 80 °C for 2 h to remove the residual solvents. Nylon films were the counter electrode. Finally, a friction nanogenerator was fabricated from the two electrodes.

2.5. Characterization. The surface morphology of mesoporous silica was characterized by a JEOL JSM-6710F field emission scanning electron microscope (Japan Electro-Optical Laboratory). Transmission electron microscopy scanning experiments were performed using

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Tecnai G2 TF20 (FEI) to detect the morphologies of mesoporous silica before and after filling perfluorooctylethanol. When the TENG is driven by a commercial linear mechanical motor, a potential difference is created between the two different friction layers, resulting in an electrical output in the external circuit. The output voltage (V_o) was measured using a NI PCIe-6259 DAQ card (National Instruments America). Short-circuit current (I_{sc}) was measured using an SR570 low-noise current amplifier (Stanford Research System).

3. RESULTS AND DISCUSSIONS

Figure 1a shows a preparation scheme for the TENG with mesoporous silica-loaded perfluorooctylethanol. The process included the preparation of mesoporous silica, loading perfluorooctylethanol into the mesopores of silica, and incorporating fluorinated mesoporous silica into acrylate. Then, we used the spin-coating method to fabricate the coating friction layer on the top surface of the Cu substrate. Lastly, the friction matching control material was fabricated with spin-coating nylon solution on the surface of PI/Cu electrode to form a PI-Cu-NY@F-SiO_2 coating/Cu TENG, in which the NY/Cu part was used as the positive friction electrode, and coating/Cu part was used as the negative friction electrode. The triboelectricity generation mechanism of the coating and nylon-based TENGs was similar to those of the other contact separation mode TENGs,[31] based on the coupling between the triboelectric effect and electrostatic induction under periodic contact and separation of the two material surfaces. When the two materials with different polarities make mechanical contacts, friction charge was present on the surface. When they were separated, the electrostatic potential difference between the two materials pushed the current to flow through the external circuit.

Figure 1b shows the SEM images of mesoporous silica. As shown in the figure, the synthesized mesoporous silicas were all spherical, with highly dispersed particles and smooth surfaces. According to the diameter distribution analysis (as shown in Figure S1), we can more intuitively know the particle size and distribution of silica. Most silica particles are around 400−500 nm. Figure 1c,d shows the transmission electron microscopy photographs of mesoporous silica before and after filling perfluorooctylethanol. Figure 1c shows that the surface of the mesoporous silica had a number of regularly ordered arrays of wormlike pore structures, with the pore size of several nanometers. Also, the pore size distributions of the mesoporous silica are shown in Figure S2; we can see that the pore size of silica is about 2 nm and the pore volume is 0.6815 cc/g. After filling perfluorooctylethanol, as shown in Figure 1d, the pore structure was difficult to observe, thereby indicating that the cavity was blocked by perfluorooctylethanol. The surface morphology of the coating TENG is shown in Figure S3. We selected a coating TENG with a 2% concentration of mesoporous silica to observe its surface morphology. We can see that the mesoporous silica is evenly distributed on the coating surface.

A working principle diagram is shown in Figure 2. When the nylon and coating rubbing layers were pressed to contact, the nylon surface was positively charged, while the coating surface was negatively charged (Figure 2b). When the nylon and coating layers were separated, the conductive Cu back electrodes generated induced charges opposite the connected friction layer due to the electrostatic induction effect and formed a potential difference. Therefore, the potential difference between the two Cu electrode resulted in the formation of current flows from the bottom Cu electrode to the top Cu electrode during the release process (Figure 2c). Once the friction charge reached an equilibrium, no current flowed through the external circuit (Figure 2d). Similarly, during the pressing process, a reverse current was detected from the top Cu electrode to the bottom Cu electrode (Figure 2e). Finally, due to the coupling of contact electrification and electrostatic induction during the pressing−releasing process of the coating TENG, the alternative electricity was generated.

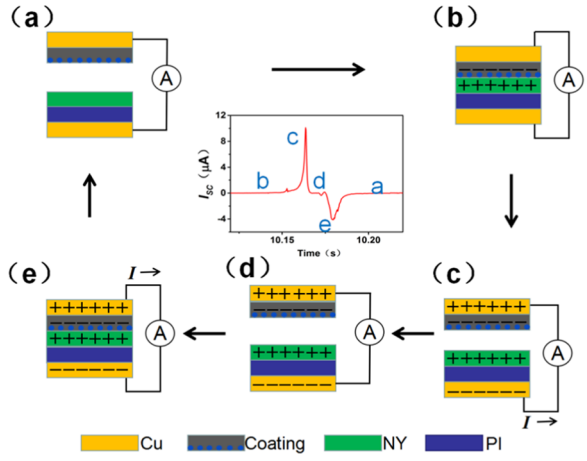


Figure 2. Scheme of working mechanisms and charge generation processes. (a) Original separated state, (b) moving to contact to generate triboelectric charges, (c) releasing to generate induced charge, the current flows from the nylon electrode to the coating electrode, (d) reaching an electrical equilibrium, no current formed, and (e) pressing to contact again, current flows from the coating electrode to the nylon electrode.

This study focuses on the achievement of coating TENG with high output performance. In our work, the pore structure of mesoporous silica was used to load perfluorooctylethanol to change the friction polarity and improve the friction output of TENG. Figure 3a−c shows I_{sc}, output voltage (V_o), and Q_c of TENGs of the pure acrylate coating, the acrylate coating with 2% mesoporous silica, and the acrylate coating with 2% mesoporous silica loaded with perfluorooctylelcohol. The coupling friction material was the nylon layer triboelectrode. After the addition of mesoporous silica modified with perfluorooctylethanol, I_{sc} and V_o of the TENG increased significantly from 2.8 to 11 μA and from 38 to 230 V, respectively. The addition of unmodified mesoporous silica caused a small increase in output performance. The output current and power of the TENG with perfluorooctylethanol is shown in Figure 3d. Stability has always been an important factor in the practical application of TENG in long-term operation. Thus, we performed a durability test of the coating TENGs with 2% mesoporous silica-loaded perfluorooctylethanol, and the results are presented in Figure 3e. The results showed that the output was extremely stable and had no tendency to decrease. As shown in Figure 3f, 48 red commercial LEDs constituting the term "LICP" were powered by TENG in series, which can be used for outdoor signs at night.

To further study the effect of the additive mesoporous SiO_2 and F-SiO_2 on the output performance of TENGs, we tested the I_{sc} and V_o performance of the coating TENGs with different silica concentrations (e.g., 0, 0.5, 1, 2, 3, and 4%), as shown in Figure 4. As shown in Figure 4a,b, different silica concentrations had significant effects on the triboelectric output. As shown in Figure 4a, the short-circuit current increased with the addition of mesoporous silica nanoparticles. However, the short-circuit

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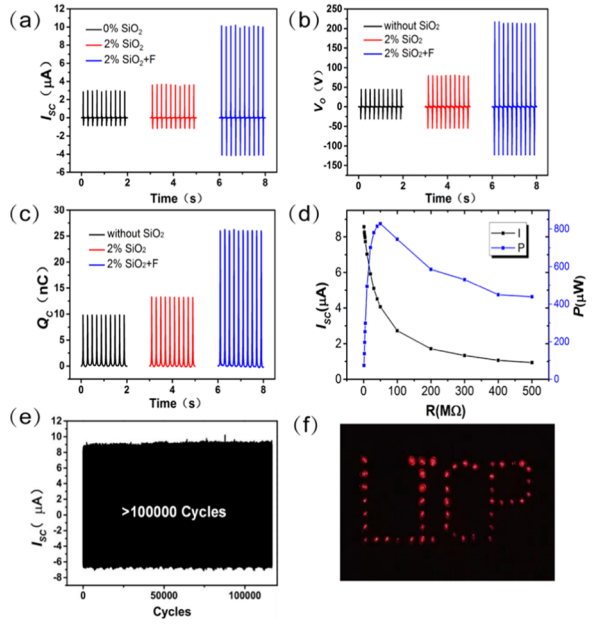


Figure 3. Performance of coating TENGs with and without F-SiO_2. (a) Short-circuit current, (b) open-circuit voltage, (c) transferred charges, (d) output current and power of the dependence of output currents, and output power of the coating TENGs containing 2% F-SiO_2 on the external loading resistance. (e) Resistance test of the coating TENGs and (f) photograph of the coating TENG lighting up commercial LEDs.

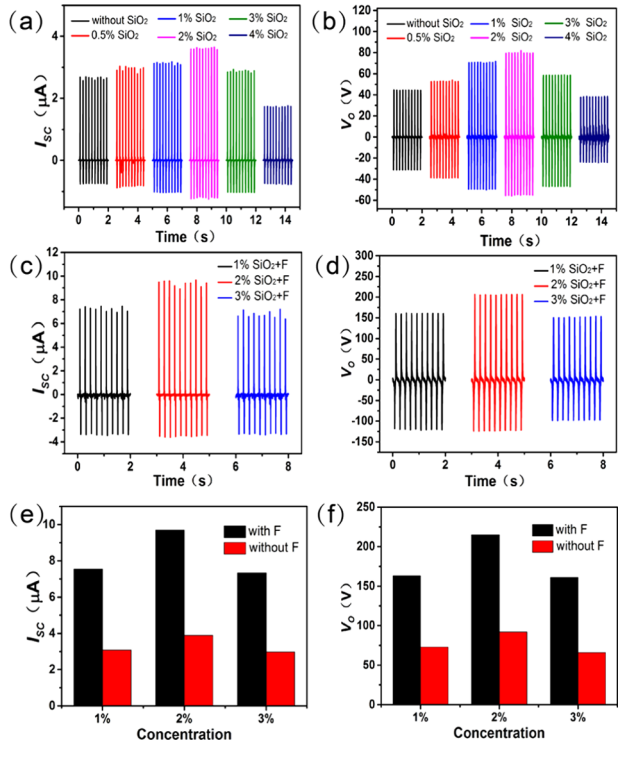


Figure 4. Performance of TENGs with different SiO_2 and F-SiO_2 concentrations. (a) Short-circuit current and (b) open-circuit voltage of coating TENG with and without pure mesoporous silica. (c) Short-circuit current and (d) open-circuit voltage of coating TENG with and without mesoporous silica-loaded perfluorooctylethanol. (e, f) Comparison of I_{sc} and V_o of coating TENGs with or without mesoporous silica loading perfluorooctylethanol.

current begins to decrease when the concentration of mesoporous silica exceeds 2%. The same trend for the V_o is also shown in Figure 4b. Silica nanoparticles increased the friction output of TENGs mainly because the silica nanoparticles had a high dielectric constant, and embedding in silica enhanced the surface charge and permittivity of the coating material to obtain an increasingly effective electrostatic induction. The reason why the increase in the dielectric constant of triboelectric materials can enhance the output performance of TENG is that the dielectric constant was related to the transferred charge density between the two friction materials. The dielectric constant increased charge density. The maximum transferred charge density σ' can be expressed as eq 1.[32−35]

$$\sigma' = \frac{\sigma_0 d_{gap}}{d_{gap} + d_{coating}/\varepsilon_{coating}} \quad (1)$$

where σ_0, d_{gap}, $d_{coating}$, and $\varepsilon_{coating}$ are the triboelectric charge densities at the equilibrium state, gap distance, thickness of the coating film, and dielectric constant of the coating film, respectively. The equation shows that the maximum transfer charge density increased as the dielectric constant increased. Although the embedded dielectric particles enhanced the surface charge density of the coating, these particles also appeared on the surface of the coating. If the concentration of silica particles is too high, the effective friction area of the coating is reduced. Therefore, the optimal concentration (2%) is the competitive balance of the two impacts. Figure 4c,d shows the influence of the concentrations of the mesoporous silica-loaded perfluorooctylethanol on the output of the coating TENGs, which has a similar trend as SiO_2 and reaches a maximum at 2%. Figure 4e,f shows a comparison of I_{sc} and V_o of the coating TENGs containing mesoporous silica with or without perfluorooctylethanol. The result shows that the loading of perfluorooctylethanol had a higher output than that of pure mesoporous silica owing to the stronger friction electronegativity of perfluorooctylethanol than that of the pure acrylate coating material.

After identifying that the mesoporous silica-loaded perfluorooctylethanol can improve the triboelectric output of TENG, we also studied the antiwear and friction-reducing effects of TENG, which was very important for its practical applications.[36−39] Figure 5 reports the friction coefficients and wear volume of the pure acrylate coating and the coating loaded with 2% mesoporous SiO_2 nanoparticle and F-SiO_2 additives. As shown in Figure 5a, for the pure acrylate, the friction coefficient rapidly increased from a small value to approximately 0.11 and remained within this range. After the addition of 2% silica nanoparticles, the time to reach the stable value was prolonged, and the stable friction coefficient of 0.045 was significantly lower than that of the pure acrylate. The friction coefficient of the coating after modification with perfluorooctylethanol was similar to that without modification, but the entire friction process was relatively stable.

Figure 5b shows the corresponding wear volume. Compared with pure acrylate, the wear volume was reduced by approximately 89% after the addition of F-SiO_2 nanoparticles. The friction coefficients for different mesoporous silica-loaded perfluorooctylethanol concentrations are shown in Figure 5c. As the fluorinated mesoporous silica concentration (1, 2, and 3%) increased, the friction coefficients gradually decreased because as the silica concentration increased, some of the silica ran to the coating surface, and the friction between the steel ball and the

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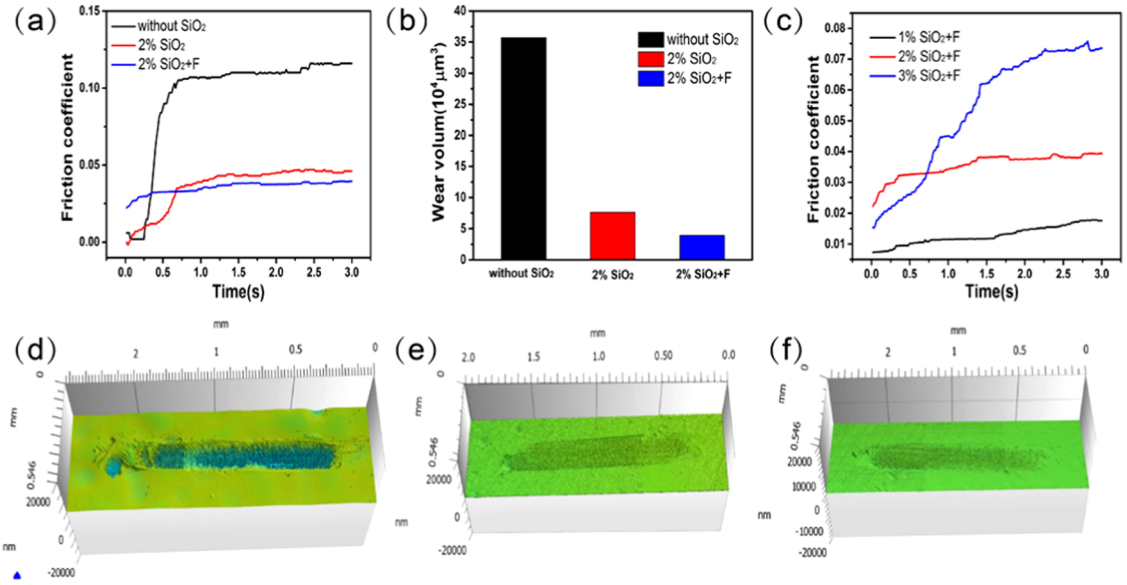


Figure 5. (a) Friction coefficients and (b) wear volumes of pure acrylate coating and the coating with 2% mesoporous SiO_2 nanoparticles and the coating with 2% mesoporous SiO_2 nanoparticle-loaded perfluorooctylethanol. (c) Friction coefficients of the coating with different mesoporous SiO_2 nanoparticle-loaded perfluorooctylethanol concentrations. (d–f) Wear scar of pure acrylate coating, coating with 2% mesoporous SiO_2 nanoparticle additive, and coating with 2% mesoporous SiO_2 nanoparticle-loaded perfluorooctylethanol.

coating at the beginning became the friction between the steel and silica balls, which decreased the friction coefficient. Figure 5d–f shows a 3D contour image of the coating after wear. The wear scar greatly decreased after the addition of silica. The results indicated that silica and F-SiO_2 nanoparticles, as an additive to TENG, exhibited excellent friction reduction and antiwear performance.

During usage, the TENG may be damaged or destroyed owing to the wear in long-term working or changes in the external environment. In particular, the perfluorooctylethanol contained in mesoporous silica spheres is easily lost due to surface damage of the coating material.[40−42] To solve this problem, we introduced mesoporous silica as the container to fill the perfluorooctylethanol and studied the healing performance of the coating TENG. To accelerate the damage of the coating, the oxygen plasma method was used to destroy the coating, which could degrade the perfluorooctylethanol on the surface of the coating. After that, heating treatment was used to transport and enrich the perfluorooctylethanol in the mesopores of silica to the outmost surface of the coating to recover the triboelectrification performance. In this experiment, we treated the surface of the coating with O_2 plasma for 3 min to destroy the coating surface and then returned the material to the oven at 60 °C for 4 h as shown in Figure 6a. Figure 6b,c shows the short-circuit currents and the contact angles of the coating TENG before and after oxygen plasma and heating treatments. As shown in Figure 6b, the initial value of the current is 9.7 μA. After plasma treatment, the current dropped to 4.5 μA. Then, the sample was placed in an oven for 4 h, and the current became 9.2 μA. The contact angle also showed a corresponding change as shown in Figure 6c. The above process was repeated three times in succession and showed that after the plasma treatment, the perfluorooctylethanol on the coating surface was destroyed. Thus, the output was reduced. After the heating treatment, the perfluorooctylethanol at the bottom of the cavity returned to the surface of the material, and the current output was restored again, indicating that the coating TENG had a healing function for energy harvesting.

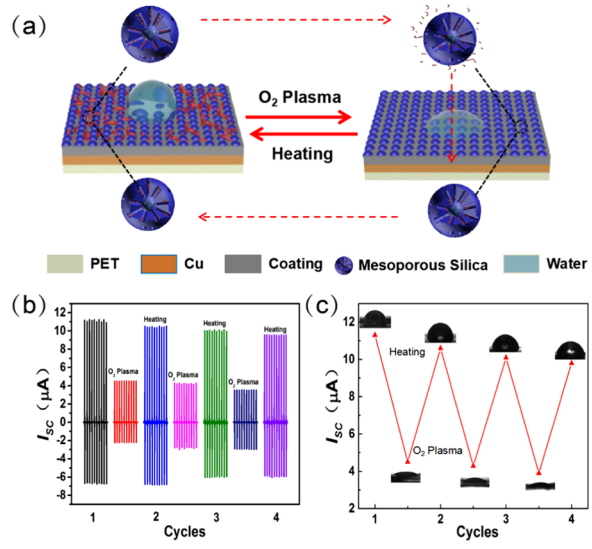


Figure 6. (a) Scheme of healing performance of the TENG friction coating with oxygen plasma damage treatment and heating self-repair process. (b, c) Short-circuit current (I_{sc}) and the contact angle of wetting behavior of TENG coating before and after oxygen plasma and heating treatments.

The coating TENG has a wide range of applications in the future. In daily life, when people go home from work, the home is sometimes dark, and fumbling for lighting is always inconvenient. At this moment, if the wall and door handle were touched gently, the light switch would light up, and the user could accurately turn the light on in the dark. In this paper, to demonstrate the application of the coating TENG in the sensing system, we sprayed the coating to the wall and door handle, designed it as a single-electrode TENG (S-TENG). Figure 7a shows a scheme of the S-TENG, which includes three parts, namely, a coating and a hand as a friction layer and copper as an electrode layer, with a small LED bulb connected to the TENG and grounded to the earth. Figure 7b shows that when a person

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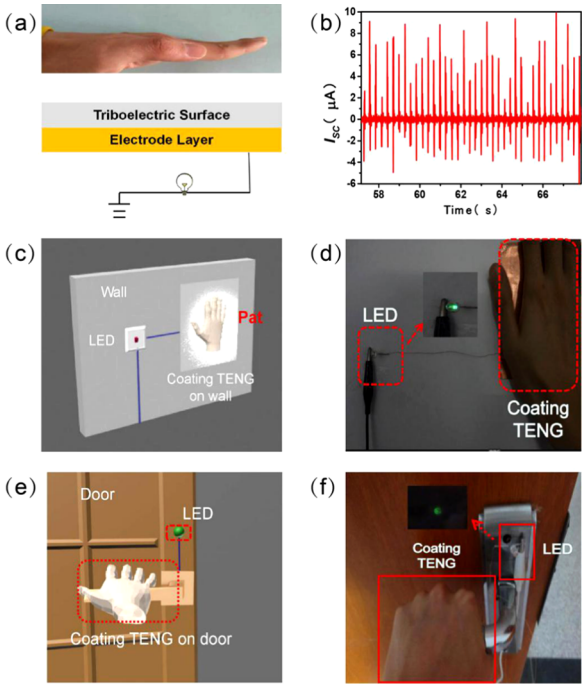


Figure 7. Applications of coating TENG. (a) Scheme of coating TENG application with a single-electrode triboelectrification mode (S-TENG). (b) Short-circuit current when a person slightly presses the S-TENG. (c) Scheme of the process of the S-TENG on the wall. (d) Image of the process of S-TENG on the wall to light LEDs by hand. (e) Scheme of S-TENG on the door. (f) Image of the process of S-TENG on the door to light LEDs by hand.

slightly presses S-TENG in a reciprocating manner from a height of approximately 5 cm by hand, the output current is approximately 8 μA, and the output shows pulse-like sharp peaks in a single direction. The scheme of the S-TENG on the wall is shown in Figure 7c. Figure 7d shows that LEDs can be used as light sources when a person presses the S-TENG slightly. Figure 7e,f shows the scheme and image of TENG spray on the door handle. Both applications showed the corresponding videos in S1 and S2, showing that the coating TENGs have potential practical applications in energy-harvesting and self-powered devices in daily life.

4. CONCLUSIONS

In summary, a new coating TENG was demonstrated that can be used for large-scale production in the future by preparing and modifying coating technology. The coating TENG consists of an acrylate coating as a friction material and was modified with mesoporous silica-loaded perfluorooctylethanol, which not only exerted the silica nanoparticles that can not only enhance antiwear and friction-reducing properties but also be utilized as a micro/nanocontainer to store F-containing substance and increase the current output. In response to the effect of changes in the external environment on TENG, a healing experiment was conducted to detect the output of TENG treated with O_2 plasma and recovery in the oven. In the past few years, although TENG has made great progress in theory and device design, its practical application has been limited. We think the combination of coating surface technology and frictional power generation is expected to become an important part of future research for its practical applications. In this paper, we applied modified acrylate coatings to friction nanogenerators and addressed some of the limitations of the device. The modified acrylate coatings contributed to the practical application of the friction nanogenerator and provided guidance for further studies on coating TENGs.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.9b22649.

SEM image and diameter distribution of the as-prepared mesoporous silica (Figure S1); pore size distributions of the mesoporous silica (Figure S2); top surface morphologies of the TENG coating electrode (Figure S3); and diameter distribution of mesoporous silica (Table S1) (PDF)

The coating TENGs have potential practical application in energy-harvesting and self-powered devices in daily life (MP4) (MP4)

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Notes

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